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# Developing smart communities

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By Belinda Salim and Chua Wen Hao, Energy Studies Institute | 31 October 2011



(Picture credit: EMA)

Promotion of smart communities is now, more than ever, crucial for Japan given the energy situation post-Fukushima, said Mr Hiroshi Asahi, Director General for Energy and Environmental Policy from Japan's Ministry of Economy, Trade and Industry. Opening the session on "Energy for Smart Communities" at the Singapore Energy Summit, he said Japan currently has four large-scale smart city demonstration projects in Yokohama, Tokyo, Keihanna and Kita-Kyushu with 50 companies taking part. A Japan Smart Communities Alliance is in place to help synergise the different industries, energy users and companies involved.

## Gas as cleanest fossil fuel

Referring to the rapidly increasing global population and urbanisation, Shell's Vice President for Gas Market Development, Mr Dick Benschop, said the development of new smart cities and retrofitting of existing cities hold the key to tackling the increased global energy demand. He added that natural gas fits nicely into the development of smart cities as it is the cleanest fossil fuel right now for power generation. Gas is also a good complement to renewable energy due to the latter's intermittency issues.

Mr Girish R Tanti, Independent Director of Suzlon Energy, said smart communities realise that energy sources are limited and hence use energy responsibly through smart technologies. Thus, strong R&D is needed to develop the renewable energy sector.

## Wind powers on

In particular, because of technological improvements, wind power is now 100 times more powerful than when it first started, and is able to compete with conventional fuels in some instances. The use of smart turbines to harness low-speed winds has also increased the potential of wind energy. In addition to technology, Mr Tanti also underscored the need for dynamic political leadership and robust public policy for renewables to play a significant role.

## Steering on to e-mobility

Turning to e-mobility, Mr Glenn Schmidt, Head of the Steering Governmental Affairs from BMW, said that in dense urban areas where range is not a big factor, electric vehicles have tremendous potential to lower carbon emissions. Based on BMW's electric vehicle test-beds in six countries, range is no longer a major concern once consumers actually start using the vehicle. He added that given the 100-120km range of most electric vehicles, they should be suitable for Singapore which is just 40km wide.

## Building blocks and stumbling blocks

Pulling the different threads together, Dr Bartosz Wojszczyk, Global Director of Smart Grid Technical Solutions, GE Energy, said that given the different elements which constitutes a smart community,

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E2Singapore

Accenture Utilities Group

List of smart energy buildings

Singapore's first zero energy building

BCA Green Mark

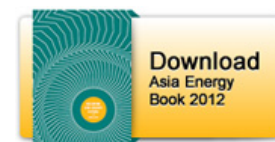
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an integrated infrastructure is critical.

Mr Michael Liebreich, Chief Executive of Bloomberg New Energy Finance, said that of the US\$243 billion worth of investments in green technology, smart communities attracts only US\$10 to US\$15 billion. Investors are unwilling to commit because it is difficult to see a return of investments from implementing smart communities. Of the 500 pilots logged by Bloomberg, most only offer partial solutions, involving too few households or only deploying simple technologies.

What was really required, according to Mr Liebreich, was a project involving millions of households and profound re-engineering such as demand management, building automation systems and changing the way mobility is being used. He added that the key focus of smart communities should be to drive energy efficiency and enable the takeup of renewables, as it is only through efficiency that investors can reap returns.

### To start anew or go with existing cities?

During the Q&A session moderated by Dr Elspeth Thomson, Senior Fellow, Energy Studies Institute, a question was raised on whether the focus should be on developing new smart cities or retrofitting existing ones. Some panelists noted that while the focus should rationally be on the design and development of new cities, aside from controlled urbanisation in China, most cities in the developing world have emerged on their own and are not planned or designed by the government, making it difficult to make them "smart".

On the other hand, it is extremely expensive to retrofit existing buildings in developed cities.

The importance to have a clear mandate from the government was also highlighted. The problem is not a technical but a social engineering one, said Mr Liebreich. Once cities set a roadmap for their future with a stated end point in mind, the appropriate technologies and solutions will find their way.

### Urbanisation as an opportunity for cooperation

The panel session also featured a video from European Commissioner for Energy Gunther Oettinger, who said that urbanisation provides a great opportunity for cooperation among governments, companies and non-profit organisations to introduce greater efficiency in urban communities. This was why the European Union launched its Smart Cities and Communities Initiative in June this year. Commissioner Oettinger hoped that Europe's example could spur the development of other smart communities around the world.

1724 reads

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